

GPU-Accelerated N-body Simulation of the Solar System

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Base Paper Overview

Base Paper: *N-body Simulations of the Solar System with CPU-based Parallel Methods*
— Tailin Zhu, School of Physics, University of Bristol (2020)

Domain: Numerical simulation of gravitational N-body systems; Solar System dynamics and parallel computing.

Governing Equation (Newtonian gravity): The acceleration of body p from all other $N - 1$ bodies is

$$\ddot{\mathbf{r}}_p = \sum_{\substack{j=1 \\ j \neq p}}^N G m_j \frac{\mathbf{r}_j - \mathbf{r}_p}{\|\mathbf{r}_j - \mathbf{r}_p\|^3}, \quad p = 1, 2, \dots, N.$$

Solved as an initial-value problem $\dot{\mathbf{y}} = f(t, \mathbf{y})$, $\mathbf{y}(t_0) = \mathbf{y}_0$, with total energy conserved (accuracy check).

Main Approach:

- Integrates the equations of motion with the second-order **velocity Verlet** method ($\Delta t = 0.1$ days).
- Compares two CPU parallel approaches: **OpenMP** (shared memory) and **MPI** (distributed memory).

Problem Statement & Project Scope

Velocity Verlet Integration (the core scheme): Positions and velocities are advanced with second-order accuracy (error $O(\Delta t^2)$):

$$\mathbf{r}_{i+1,p} = \mathbf{r}_{i,p} + \dot{\mathbf{r}}_{i,p} \Delta t + \frac{1}{2} \ddot{\mathbf{r}}_{i,p} \Delta t^2,$$
$$\dot{\mathbf{r}}_{i+1,p} = \dot{\mathbf{r}}_{i,p} + \frac{1}{2} [\ddot{\mathbf{r}}_{i,p} + \ddot{\mathbf{r}}_{i+1,p}] \Delta t.$$

Energy-conserving and stable for accelerations depending on position only — better than first-order Euler, cheaper than 4th-order Runge–Kutta.

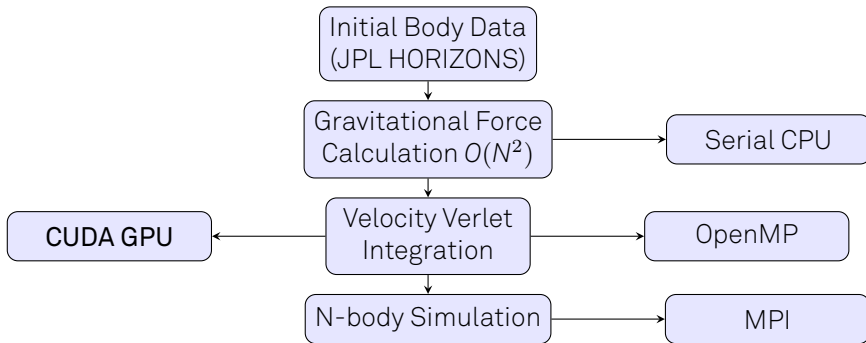
Problem Statement

- The $O(N^2)$ force evaluation makes large models expensive.
- More processors do not give proportional speed-up: communication overhead and **Amdahl's law** limit the gain.

Our Project Scope

1. Implement the N-body model with velocity Verlet; build a serial baseline.
2. Reproduce CPU parallelization with OpenMP and MPI.
3. **Extend with CUDA GPU parallelization** — the future-work direction suggested in the paper.

Expected Methodology



Experimental Comparison: Vary the number of simulated bodies; measure execution time and speed-up; compare scalability and orbit accuracy (total-energy error) across serial, OpenMP, MPI and CUDA.